COS10022

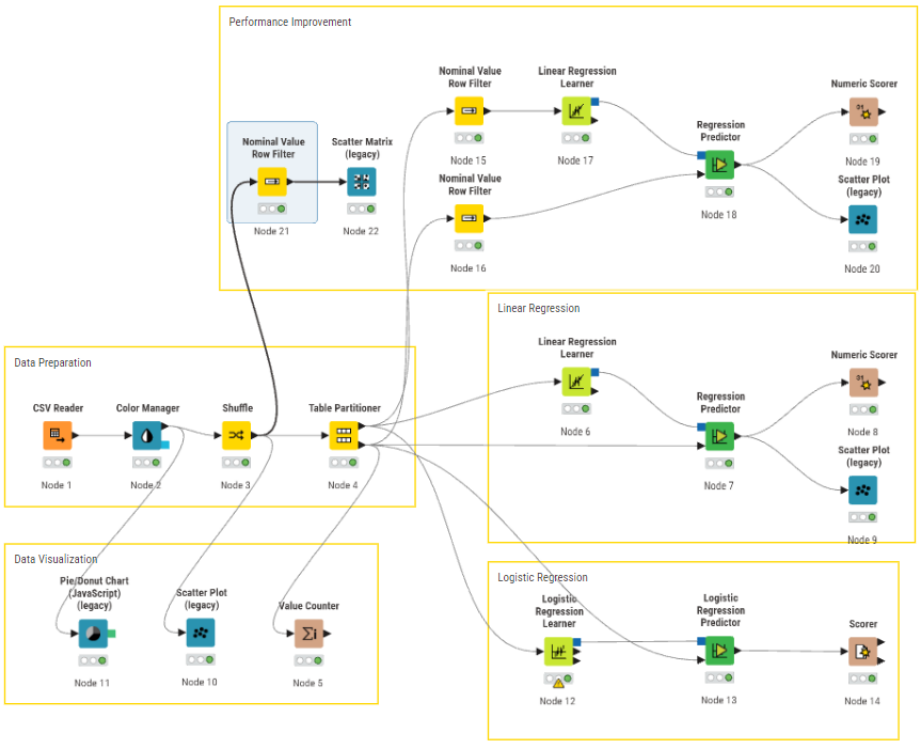
Assignment 1 template

Read and respond to each question in this template. Some questions will require you to include a screenshot of your work in KNIME – please ensure that these screenshots are clear to ensure that you get marked correctly for your work.

Step 1 – 10 marks

Follow the instructions provided to split the source data into training and test sets, then answer the following questions after splitting the data:

1. Paste a clear screenshot of the whole workflow of Assignment 1. **(7 marks)**



2. How many tuples are included in the training set? **(1 mark)**
   120 tuples

3. How many species are included in the test set? **(1 mark)**
   There are 7 species in the test set which includes of “Parkki”, “Pike”, “Whitefish”, “Bream”, “Perch” , “Roach”, and Smelt.

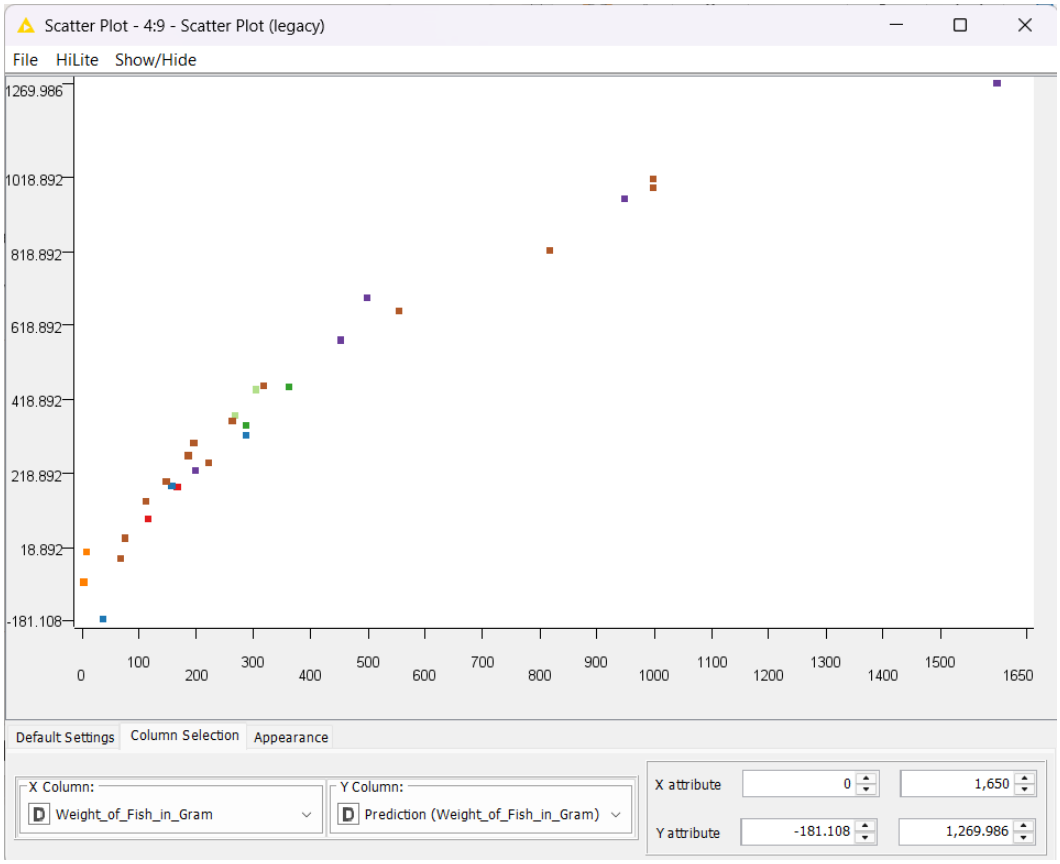
4. Do the species 'Whitefish' and 'Smelt' have the same number of tuples included in the test set? **(1 mark)**
   Yes

Step 2 – 40 marks

Build a Linear regression model using all available attributes to predict the value of the 'Weight_of_Fish_in_Gram'. Then answer the following questions after completing the model training and test.

1. What is the R^2 value of your test result? Do the species 'Whitefish' and 'Smelt' have the same number of tuples included in the test set? **(5 marks)**
   My test run gave an R^2 value of **0.918**, which basically means the model's predictions line up pretty well with the actual fish weights. Whitefish and Smelt both show up twice in the test set, so they have the same number of entries.

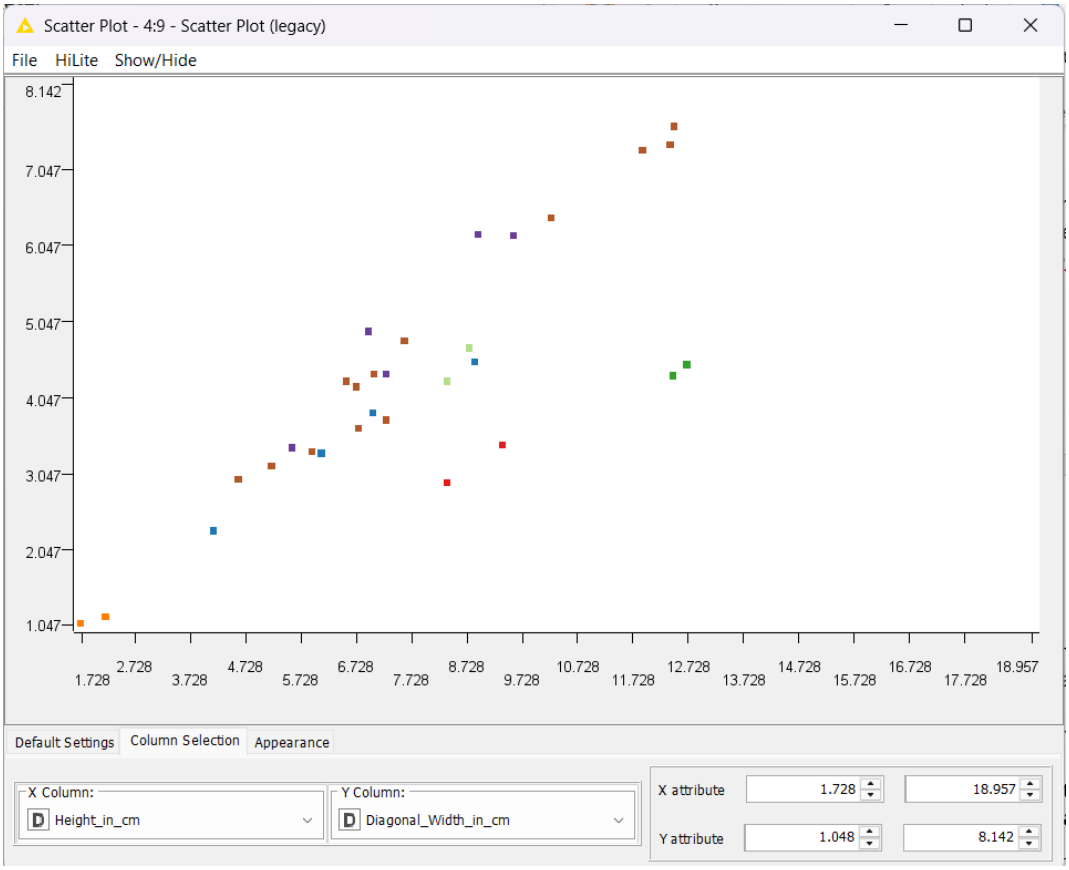
2. Provide the screenshot of the scatter plot result of your test output using 'Weight_of_Fish_in_Gram' on the x-axis **(5 marks)** and the prediction value on the y-axis **(5 marks)**. Assign different colours to the data points based on the species **(5 marks)**. **(5 + 5 + 5 = 15 marks)**



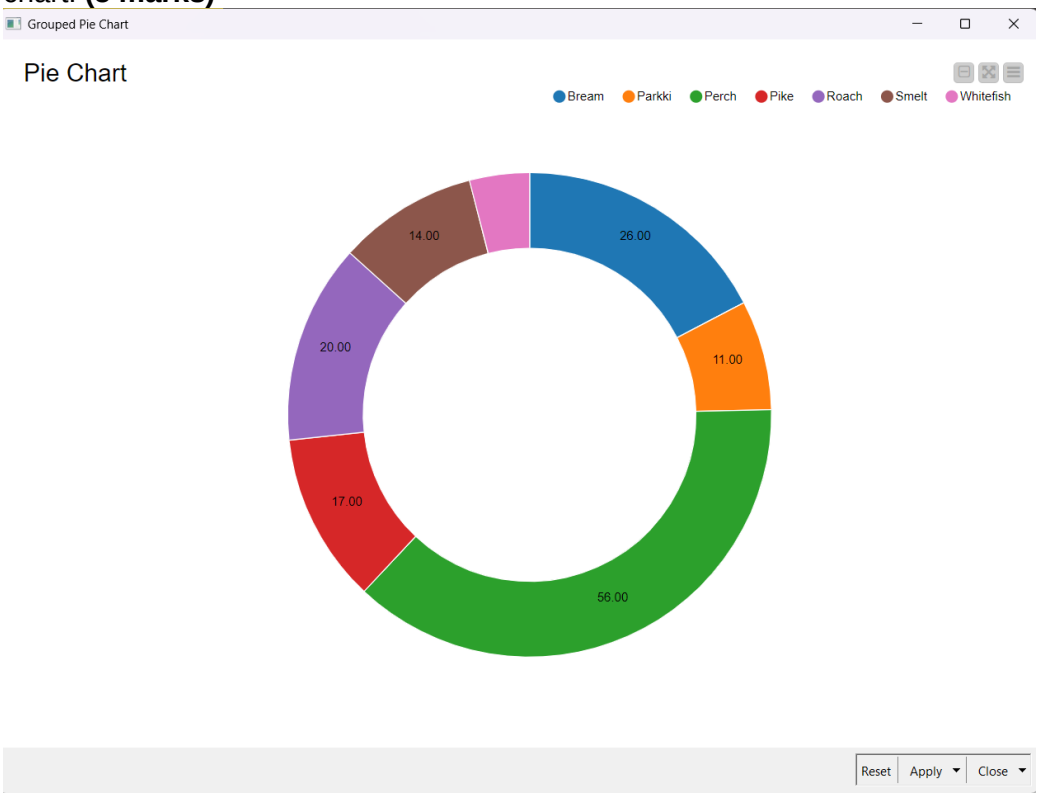
3. Which species has the heaviest predicted weight in your test results? **(5 marks)**
   Based on the prediction values generated by the model, Pike had the heaviest estimated weight in the test dataset.

4. How many prediction results are infeasible in your test result? **(5 marks)**
   3. The predicted values generated in the test data set include some that are negative; however, a negative value does not make physical sense since the variable Weight_of_Fish_in_Gram can only take positive values. This means that the negative values generated are infeasible since it is impossible for the weight of a fish to be less than zero.

5. Looking at your source data before splitting them, which two species can be easily separated from others if looking at the 'Height_in_cm' and 'Diagonal_Width_in_cm' attributes? Post your visualisation result on data observation in the report. **(5 marks)**
As observed from the scatter plot of Height_in_cm vs. Diagonal_Width_in_cm (without dividing the data set), as when I looked at the scatter plot of height vs diagonal width (before splitting the data), Whitefish and Pike stood out straight away. Both sit higher on the plot because their measurements are bigger, so they are easier to spot compared to other species. The reason is that both the species have comparatively larger values of heights and diagonals widths than the other species.



6. Draw a pie chart of the original input data before splitting it into training and test sets. Use different colours for each species and show the percentage of data in the pie chart. **(5 marks)**



Step 3 – 40 marks

Build a Logistic regression model with all attributes and use 'Smelt' as the reference category. The maximal number of epochs and epsilon should be set to **10,000** and **0.0001**, respectively. Use 3122 as the seed in the logistic regression node. Answer the following questions after completing the model training and test.

1. Which species has no True Positive (TP) case in the prediction result? **(5 marks)**
   The model produced no True Positive cases for Whitefish, meaning none of the Whitefish samples were correctly identified during testing.

2. For the species with no TP case, which species will be misplaced? **(5 marks)**
   For the species that have lacked any correct predictions, Whitefish was consistently misplaced as Perch in the test results.

3. What is the overall accuracy of the prediction result? **(5 marks)**
   The logistic regression model achieved an overall accuracy of 0.833, indicating that approximately 83.3% of the test samples were classified correctly.

4. List all species names that have 100% correctly classified test results. **(15 marks)**
   To determine which species were classified with 100% accuracy, the accuracy for each species was calculated using the standard formula:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

A species is considered 100% correctly classified only when:

- True Positives (TP) > 0
- False Positives (FP) = 0
- False Negatives (FN) = 0

This means every instance of that species was predicted correctly, and no other species were incorrectly predicted as that species.

Using the values from the classification results table:

- Pike has TP = 1, FP = 0, FN = 0 → Accuracy = 1.00
- Smelt has TP = 1, FP = 0, FN = 0 → Accuracy = 1.00

All the other species have either false positives or false negatives, so they do not reach 100% accuracy.

Therefore, the species with 100% correctly classified tests results are: Pike and Smelt.

5. Which species has a 50% chance of being misplaced into another species in the test result? **(5 marks)**
   Parkki

   FNR = 1 – Sensitivity/Recall
   = 1 – 0.5
   = 0.5 (= 50%)

6. In the test result, what percentage of the species 'Pike' is misplaced into others? **(5 marks)**
   FNR = 1 – Sensitivity/Recall
   = 1 – 1
   = 0 (0%)

Step 4 – 10 marks

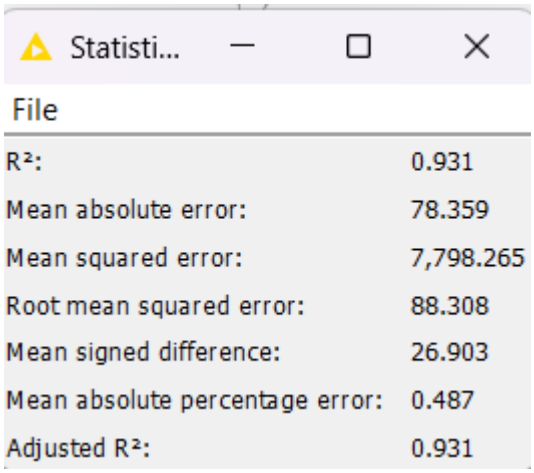
Build a new linear regression model different from the one built when working through Step 2. This time, focus on the species 'Perch' only. You are limited to using three attributes in the input to predict the 'Weight_of_Fish_in_Gram'.

Use a Scatter matrix (local) node to observe your data and decide the suitable attributes to be included. The linear regression model should be the same as the one used in Step 2, except for the input attributes. Build, train and test the model and then respond to the following questions.

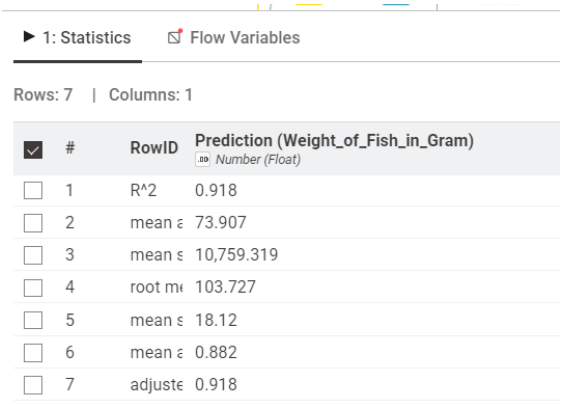
1. Give the reasons for each eliminated attribute and why they are not selected as the input. **(5 marks)**
   This is based on the data from the Scatter Matrix node, as the 2 attributes are: "Diagonal Width in cm" and "Cross_Length_in_cm"

- "Diagonal_Width_in_cm" should be removed from the list due to its weak linear correlation with Weight_of_Fish_in_Gram. This graph shows scattered points with no upward trend in it, which means that an increase in Diagonal Width does not lead to Perch weight increase. Moreover, this feature is highly correlated with some other length features, which may eventually lead to multicollinearity.
- "Cross_Length_in_cm" should be removed because of its weak correlation with weight compared to other chosen features. This graph may contain more noise and have weak linear relation, thus having little predictive value.

2. List the R^2 of your test result and compare it with the R^2 from your data analysis in Step 2. Reveal both R^2 values obtained through the modelling in Step 2 and Step 4. If you can improve the model, please demonstrate how you would do so. **(5 marks)**



Improved Model



▶ 1: Statistics Flow Variables

Rows: 7 | Columns: 1

	#	RowID	Prediction (Weight_of_Fish_in_Gram) Number (Float)
☐	1	R^2	0.918
☐	2	mean a	73.907
☐	3	mean s	10,759.319
☐	4	root m	103.727
☐	5	mean s	18.12
☐	6	mean a	0.882
☐	7	adjuste	0.918

First model

The larger the R^2 value, the greater will be the accuracy of the prediction, where in the above case, the R2 of the model improved by 0.013 upon the removal of irrelevant variables. This highlights how reduction in dimensionality through the removal of those variables that have weak correlations with the output variable can greatly improve the efficiency of the model. The regression model becomes more robust and accurate when it uses only those variables that can have a high correlation with the dependent variable.

Conclusion

Overall, this assignment gave me a good chance to try out both regression and classification in KNIME using the fish dataset. I built a linear model to estimate fish weight and a logistic model to predict species. When I focused only on Perch and picked 3 solid attributes, the regression results improved. Working through the steps helped me understand how the data behaves, how different features matter, and how tweaking the model can change the accuracy.

References:

1. KNIME (2023). KNIME. https://www.knime.com/
2. KNIME Documentation (2023). https://docs.knime.com/
3. Lane, D. (2023). Online Statistics Education. https://onlinestatbook.com/
4. NIST/SEMATECH e-Handbook of Statistical Methods (2023). https://www.itl.nist.gov/div898/handbook/

GenAI use:

I used Copilot mainly to help me understand some of the KNIME outputs and to make my explanations clearer. It also helped me rephrase a few parts so they sounded more natural. All the actual KNIME work – building the workflows, choosing the settings, and interpreting the results – was done by me. Copilot was just a support tool for checking my wording and helping me organise my answers in a structured way.